LOCAL GEOMETRY OF ALGEBRAIC STACKS

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Abstract. In this exposition, we prove an étale slice theorem for algebraic stacks following [3], showing that near a point with a linearly reductive stabiliser, stacks are étale-locally a quotient stack. This generalises the classical statement from differential geometry and Luna's analogue for affine varieties, enabling a deformation-theoretic study of the local structure of algebraic stacks.

1. Introduction

Throughout the course, the central question in our study of moduli problems that served as our guiding motivation was the following:

Question 1. Given some moduli stack $\mathcal{X}$, when does it admit a $\{$*fine*, *coarse*, *good*$\}$ moduli space?

We saw the answer to the first adjective pretty early on: $\mathcal{X}$ admits a fine moduli space if it is actually a scheme, or more precisely representable by a scheme. However, it was quickly evident that being representable is quite a restricting condition. Later, the Keel-Mori theorem provided a sufficient condition for the existence of a coarse moduli space. Whilst retaining much of the geometry of the moduli problem, it is also somewhat restrictive, requiring $\mathcal{X}$ to have finite inertia. Without this condition, algebraic stacks rarely admit coarse moduli spaces. Good moduli spaces were introduced by Jarod Alper [2] to study stacks with infinite – albeit linearly reductive – stabilisers, which one would hope still exhibit some nice geometric and uniqueness properties similar to those of GIT quotients. Adding some extra conditions, the focus of this report will then be to study the following question:

Question 2. Take $\mathcal{X}$ to be an algebraic stack over an algebraically closed field k. If $\mathcal{X}$ has linearly reductive stabilisers, does there exist an algebraic space such that $\mathcal{X} \to X$ is a good moduli space?

Recall that the main example from the course of a good moduli space was $[X/G] \to Y$, where G is a linearly reductive group acting on an affine variety X and Y a good quotient. It turns out that the study of Question 2 is tied closely to the question of whether an algebraic stack $\mathcal{X}$ is *étale-locally a quotient stack*. For instance, if $\mathcal{X}$ admits an "étale cover" of the form $[\operatorname{Spec} A/G]$, then we can locally construct the good moduli space as the morphism $[\operatorname{Spec} A/G] \to \operatorname{Spec} A^G$. One might then hope to glue these local pieces into a global good moduli space for $\mathcal{X}$. Remarkably, every "reasonable" algebraic stack with linearly reductive stabilisers is indeed a quotient stack étale-locally.

Theorem 1. *Let $\mathcal{X}$ be a quasi-separated algebraic stack which is of locally finite type over k and has affine stabilisers. Let $x \in \mathcal{X}$ be a smooth k-point with linearly reductive stabiliser group G_x.*

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Then there exists an affine and étale morphism $(U,u) \to (N_x /\!\!/ G_x, 0)$ where N_x is the normal space to x, and a cartesian diagram

$$\begin{tikzcd} ([N_x/G_x],0) \arrow[d] & ([W/G_x],w) \arrow[l] \arrow[r, "f"] \arrow[d] \arrow[dl, phantom, "\square"] & (\mathcal{X},x) \\ (N_x /\!\!/ G_x, 0) & (U,u) \arrow[l] & \end{tikzcd}$$

such that W is affine and f is étale and induces an isomorphism of stabiliser groups at w.

This is Theorem 1.2 in [3], the *étale slice theorem for algebraic stacks*. The slice theorem in differential geometry roughly states that there exists a local submanifold transversal to the orbit, or a slice, where the action of a compact Lie group is a linear action. This was extended to the action of a reductive algebraic group on an affine variety by Luna in [9]. We thus get a local answer to Question 2. Intuitively, for all smooth k-points x, $[\operatorname{Spec} A/G_x]$ is a common étale neighbourhood for $\mathcal{X}$ and $[N_x/G_x]$ (this is the étale slice), so one should expect algebraic stacks to look locally like quotient stacks near points with linearly reductive (affine) stabilisers. Furthermore, as a consequence of Theorem 1, the authors also provided sufficient and necessary conditions in answer to Question 1 for the existence of a good moduli space with some extra conditions on $\mathcal{X}$ [3, Theorem 5.3].

1.1. **Proof strategy.** The remainder of the report will be dedicated to proving this result. There are four main ingredients used in the proof of the theorem: (1) deformation theory, (2) coherent completeness, (3) Tannakian formalism, and (4) Artin approximation. In a nutshell, deformation theory is the study of a family in an infinitesimal neighbourhood of a given element. We will show that the n-th infinitesimal neighbourhood $\mathcal{N}^{[n]}$ of 0 in $\mathcal{N} := [N_x/G_x]$ is isomorphic to the n-th infinitesimal neighbourhood $\mathcal{X}_x^{[n]}$ of x in $\mathcal{X}$. The required background, such as the cotangent complex and obstruction theory, will be introduced in Section 2. Then, coherent completeness (Theorem 3.1) and Tannakian formalism (Theorem 3.2) will provide a morphism $\widehat{\mathcal{N}} \to \mathcal{X}$, where $\widehat{\mathcal{N}} := \operatorname{Spec} \widehat{\mathcal{O}}_{N,0} \times_N \mathcal{N}$. This is the f in the diagram. Finally, Artin approximation (Theorem 3.5) gives us the cartesian square and the required conditions on f. We will provide more motivation for (2)-(4) in Section 3. The full proof of Theorem 1 will be given in Section 4.

1.2. **Preliminary notations and definitions.** In this section, we state and recall some notations and definitions that would otherwise interfere with the exposition.

We fix an algebraically closed field k. Throughout, $\mathcal{X}$ will denote a *quasi-separated* algebraic stack over k, that is the diagonal map $\mathcal{X} \to \mathcal{X} \times_k \mathcal{X}$ is quasi-compact. A point $x \in |\mathcal{X}|$ is *of finite type* if there exists a representatitive $\operatorname{Spec} k \to \mathcal{X}$ of x that is locally of finite type. If $\mathcal{X}$ has *affine stabilisers*, then for all k-points x, the stabiliser group G_x is affine. Recall that a morphism $f\colon \mathcal{X} \to \mathcal{Y}$ of algebraic stacks is *cohomologically affine* if f is quasi-compact and $f_*\colon \operatorname{QCoh}(\mathcal{X}) \to \operatorname{QCoh}(\mathcal{Y})$ is exact. If $G \to S$ is an affine group scheme of finite type, then G is *linearly reductive* if $BG \to S$ is cohomologically affine. Lastly, a ring is *reduced* if it has no non-zero nilpotent elements. A scheme X is *reduced* if every local ring $\mathcal{O}_{X,x}$ is reduced, and we can define reduced stacks by usual descent yoga.

Remark 1.1. Let us reconcile this definition of linearly reductive with the usual one from lectures. If $S = \operatorname{Spec} k$, then $BG = [\operatorname{Spec} k/G]$. By setting $\mathcal{X}$ to $\operatorname{Spec} k$ and $\mathcal{F}$ to be the fibred category of quasicoherent sheaves in Proposition 3.48 of [12], we get that $\mathrm{QCoh}(BG) \simeq \mathrm{Rep}_k(G)$, the category of G-representations. Under this equivalence, a quasicoherent sheaf $\mathcal{I}$ on BG corresponds to a G-module V. The pushforward f_* corresponds to taking G-invariants, and the condition that f_* is exact means that the functor $V \mapsto V^G$ from G-representations to vector spaces is exact. If $0 \to V_1 \to V_2 \to V_3 \to 0$ is an exact sequence of G-dimensional representations (which we may assume to be finite dimensional [1, Proposition 2.5]), applying the functor $\mathrm{Hom}(V_3, -) = \mathrm{Hom}(k, V_3^\vee \otimes -) = (V_3^\vee \otimes -)^G$ is exact, so the sequence splits. This means that $\mathrm{Rep}_k(G)$ is completely reducible, which is equivalent to G being linearly reductive in the classical sense.

2. Deformation theory

As mentioned in the introduction, deformation theory is the study of the local geometry near a point. By understanding how a family behaves locally, one might wish to extend this neighbourhood globally whilst retaining all its local geometric properties. We will see later that this is typically met with obstructions.

2.1. **Residual gerbes.**

To begin, we introduce the residual gerbe. Just like how residue fields are the "smallest" piece of structure of a point in a scheme, residual gerbes is the smallest "stacky piece".

Definition 2.1. Let x be a point of $\mathcal{X}$. The *residual gerbe* $\mathcal{G}_x$ *at* x is the unique strictly full subcategory of $\mathcal{X}$ such that $\mathcal{G}_x$ is a reduced, locally Noetherian algebraic stack and $|\mathcal{G}_x|$ is a singleton which maps to x via the map $|\mathcal{G}_x| \to |\mathcal{X}|$.

As $\mathcal{X}$ is quasi-separated, any point $\operatorname{Spec} k \to \mathcal{X}$ is automatically quasi-compact, so $\mathcal{G}_x \to \mathcal{X}$ is a locally closed immersion. By definition, any point in $\mathcal{X}$ is of finite type, so $\mathcal{G}_x$ exists ([5, 06G3]). It is also unique ([5, 06MT]), and $\mathcal{G}_x$ is an actual gerbe ([5, 06QK]), so it makes sense to call $\mathcal{G}_x$ *the* residual *gerbe* in this context. The following lemmas will be useful in reducing Theorem 1, with the first one included for completeness.

Lemma 2.2 ([5, 06MP])**.** *There exists a locally finitely presented, surjective, flat morphism* $\operatorname{Spec} k \to \mathcal{G}_x$ *where* k *is a field.*

Proof. Let W be a nonempty affine scheme and let $W \to \mathcal{G}_x$ be a smooth morphism. Let $w \in W$ be a closed point and set $k = \kappa(w)$. The morphism $w\colon \operatorname{Spec} k \to W$ is of finite presentation since W is locally noetherian [5, 01TV]. It is also flat as W is reduced [5, 033B], so $\mathcal{O}_{W,w} = k$. This means that the composition $\operatorname{Spec} k \xrightarrow{w} W \to \mathcal{G}_x$ is locally of finite type as smooth morphisms are locally of finite type [5, 06MH], and compositions of locally finite type morphisms is locally of finite type [5, 03XG]. It is also surjective since $|\mathcal{G}_x|$ is a singleton, and flat as composition of flat morphisms are flat. □

Lemma 2.3. *Assume that* $\mathcal{X}$ *is additionally noetherian. If* $x \in |\mathcal{X}|$ *and* G_x *is a smooth affine stabiliser of* x*, then* $\mathcal{G}_x \simeq BG_x$*.*

Proof. We will show that there is a monomorphism from the prestack BG_x^{pre} to $\mathcal{X}$ (viewed as a prestack). From lectures, $BG_x^{\mathrm{pre}}(T) = [k(T)/G_x(T)]$ over a k-scheme T, so this unique object gets mapped to $T \to \operatorname{Spec} k \xrightarrow{x} \mathcal{X}$. Furthermore, a morphism $T' \to T$ corresponds to a map $T' \to G_x$, thus showing that $BG_x^{\mathrm{pre}} \to \mathcal{X}$ is a monomorphism. By stackification, monomorphisms are preserved so $BG_x \to \mathcal{X}$ is a monomorphism. Now $\operatorname{Spec} k$ is reduced as k is a field and G_x is reduced since it is smooth [10, Prop. 1.18], so BG_x is reduced by smooth descent. It is also trivially locally noetherian since it is covered $\operatorname{Spec} k$. As the residual gerbe is unique, $\mathcal{G}_x \simeq BG_x$. Note that the condition on G_x being a smooth and affine stabiliser is required in the definition of a classifying stack. □

2.2. **Infinitesimal neighbourhoods.** Given the motivation at the start of this section, let us define precisely what we mean by an infinitesimal neighbourhood.

Definition 2.4. If $\mathcal{Z} \subseteq \mathcal{X}$ is a closed substack defined by the ideal sheaf $\mathcal{J} \subseteq \mathcal{O}_{\mathcal{X}}$, the *n-th nilpotent thickening* of $\mathcal{Z}$ is the closed stack $\mathcal{X}_{\mathcal{Z}}^{[n]} \to \mathcal{X}$ defined by $\mathcal{J}^{n+1}$.

Definition 2.5. Let $x \in |\mathcal{X}|$ be a closed point. The *n-th infinitesimal neighbourhood of* x is the closed substack $\mathcal{X}_x^{[n]} \to \mathcal{X}$ defined by the n-th nilpotent thickening of the residual gerbe $\mathcal{G}_x \to \mathcal{X}$.

If $f\colon (\mathcal{X}, x) \to (\mathcal{Y}, y)$ is a morphism, denote $f^{[n]}\colon \mathcal{X}_x^{[n]} \to \mathcal{Y}_y^{[n]}$ to be the induced map on the n-th infinitesimal neighbourhoods. We will also let $\mathfrak{m}_x \subset \mathcal{O}_{\mathcal{X}}$ denote the ideal sheaf defining $\mathcal{X}_x^{[0]}$.

Thickenings also have the property that their underlying topological spaces are equal [5, 04EX]. This is a consequence of the following lemma.

Lemma 2.6. *Let A be a ring, $\mathfrak{m} \subseteq A$ a nilpotent ideal, and $M \to N$ a homomorphism of A-modules. If N is flat over A and $M/\mathfrak{m}M \to N/\mathfrak{m}N$ is an isomorphism, so is $M \to N$.*

Proof. Let $C \coloneqq \operatorname{coker}(M \to N)$. Then $C/\mathfrak{m}C = (\operatorname{coker} M/\mathfrak{m}M \to N/\mathfrak{m}N) = 0$ since $M/\mathfrak{m}M \to N/\mathfrak{m}N$ is an isomorphism. This means that $C = \mathfrak{m}C$. Since $\mathfrak{m}$ is nilpotent, there exists some n such that $\mathfrak{m}^n = 0$, which means that $C = \mathfrak{m}^n C = 0$, so $\operatorname{coker}(M \to N) = 0$. Similarly, let $K \coloneqq \ker(M \to N)$. As N is flat, tensoring with $A/\mathfrak{m}$ preserved exactness, so $K/\mathfrak{m}K = \ker(M/\mathfrak{m}M \to N/\mathfrak{m}N) = 0$. Again, $K = \mathfrak{m}^n K = 0$, so $M \to N$ is an isomorphism. □

In fact, more could be said if we know the first-order thickenings of neighbourhoods are the same.

Lemma 2.7. *Let $f\colon (\mathcal{X}, x) \to (\mathcal{Y}, y)$ be a morphism of noetherian stacks with the assumption that $|\mathcal{X}| = \{x\}$ and $|\mathcal{Y}| = \{y\}$ contain only one point. Furthermore, assume that $f^{-1}(\mathcal{Y}_y^{[0]}) \simeq \mathcal{X}_x^{[0]}$.*

(1) If $f^{[1]}$ is a closed immersion, then $f^{[n]}$ is a closed immersion for all n.

(2) If $f^{[1]}$ is a closed immersion and there is an isomorphism

$$\bigoplus_{n \geq 0} \mathfrak{m}_x^n/\mathfrak{m}_x^{n+1} \simeq (f^{[0]})^* \bigoplus_{n \geq 0} \mathfrak{m}_y^n/\mathfrak{m}_y^{n+1}$$

of graded $\mathcal{O}_{\mathcal{X}_x^{[0]}}$-modules, then $f^{[n]}$ is an isomorphism for all $n \geq 0$.

Proof. We may assume that $f\colon \operatorname{Spec} A \to \operatorname{Spec} B$ is a morphism of local schemes. The condition that $f^{[1]}$ is a closed immersion is equivalent to $B/\mathfrak{m}_B^2 \to A/\mathfrak{m}_A^2$ being surjective. For (1), we need to show that if $B/\mathfrak{m}_B^2 \to A/\mathfrak{m}_A^2$ is surjective, then $B \to A$ is surjective. We claim that $\mathfrak{m}_B A \hookrightarrow \mathfrak{m}_A$ is surjective, that is the image of $\mathfrak{m}_B$ in A is all of $\mathfrak{m}_A$. By Nakayama's lemma [5, 00DV(6)], it suffices to show that $\mathfrak{m}_B A/\mathfrak{m}_A\mathfrak{m}_B A \to \mathfrak{m}_A/\mathfrak{m}_A\mathfrak{m}_A = \mathfrak{m}_A/\mathfrak{m}_A^2$ is surjective. By the hypothesis, the composition $\mathfrak{m}_A/\mathfrak{m}_A^2 \to \mathfrak{m}_B A/\mathfrak{m}_A\mathfrak{m}_B A \to \mathfrak{m}_A/\mathfrak{m}_A^2$ is surjective, so the second map is indeed surjective. Since $B/\mathfrak{m}_B \simeq A/\mathfrak{m}_A = A/\mathfrak{m}_B A$, we get that $B/\mathfrak{m}_B \to A/\mathfrak{m}_B A$ is surjective, so another application of Nakayama's lemma gives that $B \to A$ is surjective which proves part (1).

The isomorphism of graded $\mathcal{O}_{\mathcal{X}_x^{[0]}}$-modules implies the dimensions (over k) of the same graded pieces $\mathfrak{m}_B^n/\mathfrak{m}_B^{n+1}$ and $\mathfrak{m}_A^n/\mathfrak{m}_A^{n+1}$ are equal. This means that the maps $\mathfrak{m}_B^n/\mathfrak{m}_B^{n+1} \to \mathfrak{m}_A^n/\mathfrak{m}_A^{n+1}$ induced by f, which are surjective by part (1), are isomorphic, hence $f^{[n]}$ is an isomorphism for all n, proving part (2). □

On the level of rings, the "expansion" of a ring is defined by quotients whose kernel is a nilpotent ideal, hence the name nilpotent thickening. In particular, if the product of two elements in the kernel is zero, we get the following definition.

Definition 2.8. A *square-zero extension*, or *first-order thickening*, of an algebraic stack $\mathcal{X}$ by a quasi-coherent sheaf $I \in \mathrm{QCoh}(\mathcal{X})$ is a closed immersion $i\colon \mathcal{X} \to \mathcal{X}'$ such that

$$i_*(I) \simeq \ker(\mathcal{O}_{\mathcal{X}'} \to i_*(\mathcal{O}_{\mathcal{X}}).$$

The isomorphism in the definition implies that (the pushforward of) I^2 is the zero sheaf on $\mathcal{X}'$, so $\mathcal{X}'$ is a nilpotent thickening of $\mathcal{X}$.

2.3. **The cotangent complex.** An important part of the proof of Theorem 1 is to show that there exists an isomorphism between the n-th infinitesimal neighbourhoods of x in $\mathcal{X}$ and 0 in $\mathcal{N}$. To do so, we need to introduce the *cotangent complex* associated to a morphism of algebraic stacks. We will not give an explicit definition as it falls out of the scope of this report, but rather give some properties that will be required in the proof. Before we do so, let us discuss some homological algebra.

Recall that a *Grothendieck category* is an abelian category with a generator, direct sums, and where all filtered colimits are exact. The category $\mathrm{QCoh}(\mathcal{X})$ is Grothendieck ([5, 0781]), and thus has enough injectives ([5, 079H]), meaning we can do homological algebra! Let us introduce another type of complex that will be used in defining right-derived functors on the unbounded derived category. Denote by $K(\mathcal{A})$ the homotopy category of an abelian category $\mathcal{A}$.

Definition 2.9. A complex $I^\bullet$ is *K-injective* if we have $\mathrm{Hom}_{K(\mathcal{A})}(M^\bullet, I^\bullet) = 0$ whenever $M^\bullet$ has trivial cohomology.

Definition 2.10. The *unbounded derived category* $D(\mathrm{QCoh}(\mathcal{X}))$ is the homotopy category of K-injective complexes in $\mathrm{QCoh}(\mathcal{X})$.

Remark 2.11. Any complex $M^\bullet$ in a Grothendieck category is quasi-isomorphic to a K-injective complex $I^\bullet$ ([5, 079P]), so $M^\bullet \to I^\bullet$ is in a sense an "injective resolution" of $M^\bullet$ in $D(\mathrm{QCoh}(\mathcal{X}))$.

We can define the *derived push-forward* $Rf_*\colon D(\mathrm{QCoh}(\mathcal{X})) \to D(\mathrm{QCoh}(\mathcal{Y}))$ of a morphism $f\colon \mathcal{X} \to \mathcal{Y}$, which applies f_* to a K-injective complex. The *derived pull-back* $Lf^*\colon D(\mathrm{QCoh}(\mathcal{Y})) \to D(\mathrm{QCoh}(\mathcal{X}))$ will be defined as the left adjoint of Rf_*, which exists by [5, 07BD].

Some properties characterising the cotangent complex are listed below.

Theorem 2.12 ([8, 9.1.2])**.** *For a morphism $f\colon \mathcal{X} \to \mathcal{Y}$ of algebraic stacks, there is a canonical* (relative) cotangent complex

$$\mathbb{L}_{\mathcal{X}/\mathcal{Y}} \in D(\mathrm{QCoh}(\mathcal{X}))$$

such that the following hold:

(1) If f is locally of finite presentation, for any smooth morphism $\operatorname{Spec} A \to \mathcal{X}$, the pullback of $\mathbb{L}_{\mathcal{X}/\mathcal{Y}}$ to $\operatorname{Spec} A$ is quasi-isomorphic to a bounded complex of finite free A-modules.

(2) The relative cotangent complex is supported in cohomological degree less than or equal to 1, that is, $\mathbb{L}_{\mathcal{X}/\mathcal{Y}} \in D(\mathrm{QCoh}(\mathcal{X}))^{\leq 1}$. Furthermore, if f is representable, then $\mathbb{L}_{\mathcal{X}/\mathcal{Y}} \in D(\mathrm{QCoh}(\mathcal{X}))^{\leq 0}$.

(3) If f is a finitely presented, smooth, not necessarily representable morphism of algebraic stacks, then $\mathbb{L}_{\mathcal{X}/\mathcal{Y}}$ is supported in degree 0 and 1, and is locally quasi-isomorphic to a bounded complex of projective modules.

(4) If f is representable by algebraic spaces, then there is a canonical morphism $\mathbb{L}_{\mathcal{X}/\mathcal{Y}} \to \Omega_{\mathcal{X}/\mathcal{Y}}$ that induces and isomorphism $H^0(\mathbb{L}_{\mathcal{X}/\mathcal{Y}}) \simeq \Omega_{\mathcal{X}/\mathcal{Y}}$, where $\Omega_{\mathcal{X}/\mathcal{Y}}$ is the sheaf of Kähler differentials[1]. If f is smooth, then $\mathbb{L}_{\mathcal{X}/\mathcal{Y}} \simeq \Omega_{\mathcal{X}/\mathcal{Y}}$.

(5) If $\mathcal{X} \xrightarrow{f} \mathcal{Y} \xrightarrow{g} \mathcal{Z}$ is a composition of morphisms, then there exists a distinguished triangle

$$Lf^*\mathbb{L}_{\mathcal{Y}/\mathcal{Z}} \xrightarrow{Df} \mathbb{L}_{\mathcal{X}/\mathcal{Z}} \longrightarrow \mathbb{L}_{\mathcal{X}/\mathcal{Y}} \longrightarrow \left(Lf^*\mathbb{L}_{\mathcal{Y}/\mathcal{Z}}\right)[1].$$

We say that $\mathbb{L}_{\mathcal{X}/\mathcal{Y}}$ is a *perfect complex* if it satisfies the second part of (3).

2.4. **Obstructions.** As hinted before, it is not always simple to expand the neighbourhoods we are looking at. The cotangent complex allows us to empirically describe the deformations and obstructions of “singularities”, given by the following theorem.

Theorem 2.13 ([11, Theorem 1.5])**.** *Consider the commutative diagram*

$$\begin{tikzcd}
\mathcal{X} \arrow[rr, "i"] \arrow[dr, "f"] \arrow[ddr] & & \mathcal{X}' \arrow[ddr] & \\
& \mathcal{Y} \arrow[rr, "j"] \arrow[d] & & \mathcal{Y}' \arrow[d] \\
& \mathcal{Z} \arrow[rr] & & \mathcal{Z}'
\end{tikzcd}$$

of algebraic stacks, where the horizontal maps are square-zero extensions. Let $I = \ker(\mathcal{O}_{\mathcal{X}'} \to i_*(\mathcal{O}_{\mathcal{X}})) \in \mathrm{QCoh}(\mathcal{X})$. Then there is a canonical obstruction class

$$\mathrm{ob}(f,i,j) \in \mathrm{Ext}^1(Lf^*\mathbb{L}_{\mathcal{Y}/\mathcal{Z}}, I) \coloneqq \mathrm{Hom}(Lf^*\mathbb{L}_{\mathcal{Y}/\mathcal{Z}}, I[1])$$

such that the diagram has a filling $\mathcal{X}' \to \mathcal{Y}'$ if and only if $\mathrm{ob}(f,i,j) = 0$.

We will use this result to show that there exist lifts between infinitesimal neighbourhoods.

[1]The sheaf of Kähler differentials for a morphism of stacks $f\colon \mathcal{X} \to \mathcal{Y}$ representable by algebraic spaces is defined as the differentials of the morphism $X \to \operatorname{Spec} A$, where X is the scheme (or algebraic space) representing $\mathcal{X}$ and $\mathcal{Y} = \operatorname{Spec} A$ is a reduction by smooth descent.

3. From local to global

In this section, we state some results needed relating to *effectivisation* and *algebraisation*. The former refers to realising a compatible system of infinitesimal data as coming from a formal object over a larger base. Here, formal refers to the construction of an object as a limit of thickenings. The second condition refers to passing from a formal object to an actual algebraic object. This is a common technique used in deformation and moduli theory where one starts off with some infinitesimal data, work formally to study the geometry near a point with all its thickenings, then ask whether the solution algebraises.

3.1. **Effectivisation.**

The prototypical example of effectivisation is that of formal GAGA [6, III_1, 5.1.4], which provides an equivalence of categories

$$\mathrm{Coh}(X) \to \varprojlim \mathrm{Coh}(X_n),$$

where X is a proper scheme of a complete local noetherian ring $(A, \mathfrak{m})$ and $X_n := X \times_A A/\mathfrak{m}^{n+1}$. This result allows one to "effectivise" formal deformations, that is to promote a compatible family of sheaves over all infinitesimal neighborhoods to a single sheaf on X. The following theorem is a more refined version.

Theorem 3.1 (Coherent completeness, [3, Theorem 1.3])**.** *Let G be a linearly reductive affine group scheme over k acting on a noetherian affine scheme* $\mathrm{Spec}\, A$*. Suppose $x \in \mathrm{Spec}\, A$ is a k-point fixed by G and that A^G is a complete local ring. If $\mathcal{X} = [\mathrm{Spec}\, A/G]$ and $\mathcal{X}_x^{[n]}$ is the n-th infinitesimal neighbourhood of x, then the natural functor*

$$\mathrm{Coh}(\mathcal{X}) \to \varprojlim_n \mathrm{Coh}(\mathcal{X}_x^{[n]})$$

is an equivalence of categories.

The following result is inspired by Tannaka duality, which asserts that one can recover a compact topological group from its category of representations.

Theorem 3.2 (Tannakian formalism, [7, Theorem 1.1])**.** *Let $\mathcal{X}$ be an excellent stack and $\mathcal{Y}$ be a noetherian algebraic stack with affine stabilisers. Then the natural functor*

$$\mathrm{Hom}(\mathcal{X}, \mathcal{Y}) \to \mathrm{Hom}_{r\otimes,\simeq}(\mathrm{Coh}(\mathcal{Y}), \mathrm{Coh}(\mathcal{X}))$$

is an equivalence of categories, where $\mathrm{Hom}_{r\otimes,\simeq}(\mathrm{Coh}(\mathcal{Y}), \mathrm{Coh}(\mathcal{X}))$ *is the category whose objects are right-exact monoidal functors* $\mathrm{Coh}(\mathcal{Y}) \to \mathrm{Coh}(\mathcal{X})$ *and morphisms are natural isomorphisms of functors.*

This theorem says that morphisms between stacks and can be completely characterised by right-exact monoidal functors between their categories of coherent sheaves. This is a categorical incarnation of the principle that "a space is determined by its sheaf theory", generalising classical Tannaka duality to the stacky setting.

For our purposes, we will not define what it means for $\mathcal{X}$ to be an excellent stack, but its application in the proof of Theorem 1 will satisfy this technical condition.

3.2. **Algebraisation.** Recall the definition of a complete ring, which encodes the "infinitesimal neighborhood" of a ring R at an ideal I.

Definition 3.3. Let R be a ring and I an ideal. The *completion of R with respect to I*, or the *I-adic completion of R*, is defined to be the limit

$$\widehat{R}_I := \varprojlim R/I^n.$$

If R is a local ring, for example the ring of functions at a point, we take the completion of R with respect to the maximal ideal and drop the subscript.

To motivate Theorem 3.5, consider a noetherian ring A and $\mathfrak{m} \subseteq A$ an ideal. If two "algebraic structures" S and S' over a noetherian ring A induce isomorphic structures on $\widehat{A}_{\mathfrak{m}}$, then we wish to know if S is actually isomorphic to S'. The algebraic structures in consideration will be classified by a functor

$$F\colon A\text{-algebras} \to \mathcal{S}ets,$$

which takes an A-algebra to the set of isomorphism classes of structures over that algebra. By functoriality, an element $\xi \in F(A)$ or $\widehat{\xi} \in F(\widehat{A}_{\mathfrak{m}})$ induces an element of $F(A/\mathfrak{m}^N)$ for all integers N. We wish to determine when ξ and $\widehat{\xi}$ agree on $F(A/\mathfrak{m}^N)$, which will be our étale neighbourhoods. First, we would like to consider functors which have a "finiteness" condition.

Definition 3.4. Let S be a scheme. A contravariant functor $F\colon (\mathcal{S}ch/S) \to \mathcal{S}ets$ is *limit-preserving* or *locally of finite presentation* if for every affine scheme T over S which is a limit $T = \varinjlim T_i$ of affine schemes T_i over S, we have

$$F(T) = \varinjlim F(T_i).$$

For example, the functor h_X representing X is limit preserving if and only if X is locally of finite presentation over S ([5, 01ZC]). We now state the approximation theorem.

Theorem 3.5 (Artin approximation [4, Cor. 2.2])**.** *Let S be a scheme of finite type over k and let*

$$F\colon \mathcal{S}ch/S \to \mathcal{S}ets$$

be a limit-preserving contravariant functor. Let $s \in S$ be a k-point and $\widehat{\xi} \in F(\operatorname{Spec} \widehat{\mathcal{O}}_{S,s})$. Then there exists an étale morphism $(S', s') \to (S, s)$ and an element $\xi' \in F(S')$ such that the restrictions of $\widehat{\xi}$ and ξ' to $\operatorname{Spec}(\mathcal{O}_{S,s}/\mathfrak{m}_s^{N+1})$ coincide (under the identification $\mathcal{O}_{S,s}/\mathfrak{m}_s^{N+1} \simeq \mathcal{O}_{S',s'}/\mathfrak{m}_{s'}^{N+1}$) for any integer $N \geq 0$.

Intuitively, this shows that even if we don't fully understand the behaviour of the completed object, it can always be approximated by algebraic objects. The theorem tells us that this approximation eventually stabilises: after finitely many steps in the construction of the completion – each of which involves algebraic objects – we reach one that actually comes from the completed object itself in some étale neighbourhood.

4. Proof of Theorem 1

We finally have enough machinery to prove Theorem 1.

Proof of Theorem 1. Since $x \in |\mathcal{X}|$ is locally closed (see 2.1), we can assume that x is a closed point by replacing $\mathcal{X}$ by an open substack. Let $\mathfrak{m}_x$ be the sheaf of ideals defining x and $N_x \coloneqq (\mathfrak{m}_x/\mathfrak{m}_x^2)^\vee$ be the normal space at x, which we view as an affine scheme under the identification $N_x = \operatorname{Spec}\operatorname{Sym}^\bullet N_x^\vee$. Define the quotient stack $\mathcal{N} \coloneqq [N_x/G_x]$ and let $\mathcal{N}^{[n]}$ be the n-th infinitesimal neighbourhood of 0.

Since G_x is linearly reductive, we claim that there are compatible isomorphisms $\mathcal{X}_x^{[n]} \to \mathcal{N}^{[n]}$. First, we show by induction that there is a lift $\mathcal{X}_x^{[0]}$ to a unique morphism $t_n\colon \mathcal{X}_x^{[n]} \to \mathcal{G}_x$ for all n. Here, $\mathcal{X}_x^{[0]} = \mathcal{G}_x = BG_x$ by Lemma 2.3. Since the base case is trivial, it suffices to find a lift from $t_n\colon \mathcal{X}_x^{[n]} \to BG_x$ to $t_{n+1}\colon \mathcal{X}_x^{[n+1]} \to BG_x$ which would provide a lift from $\mathcal{X}_x^{[0]}$ by the induction hypothesis. By Lemma 2.2, we are therefore in the following situation:

$$\begin{tikzcd}
BG_x \arrow[rr] \arrow[dr, "\mathrm{id}"] \arrow[ddr] & & \mathcal{X}_x^{[n]} \arrow[ddr, "t_n"] & \\
& BG_x \arrow[rr] \arrow[d] & & \mathcal{X}_x^{[n+1]} \arrow[d, "t_{n+1}"] \\
& \operatorname{Spec} k \arrow[rr] & & BG_x
\end{tikzcd}$$

The obstruction to the lift is an element of the group $\operatorname{Ext}^1(\mathbb{L}_{BG_x/k}, \mathfrak{m}_x^{n+1}/\mathfrak{m}_x^{n+2})$ by Theorem 2.13. Since $BG_x \to \operatorname{Spec} k$ is smooth, $\mathbb{L}_{BG_x/k}$ is perfect and supported in degrees zero and one by Theorem 2.12(3). As G_x is linearly reductive, the quotient stack $BG_x \to \operatorname{Spec} k$ is cohomologically affine. This means that $\operatorname{Ext}^1(\mathbb{L}_{BG_x/k}, \mathfrak{m}_x^{n+1}/\mathfrak{m}_x^{n+2}) = 0$ since taking global sections kills higher Ext groups, so there exists a filling for the diagram.

Importantly, this means that $\mathcal{X}_x^{[1]}$ factors through the identity map of $\mathcal{X}_x^{[0]}$. This implies that $\mathcal{X}_x^{[1]} \simeq \mathcal{N}^{[1]}$ as they are defined by the same coherent sheaf $\mathfrak{m}_x/\mathfrak{m}_x^2$ by Lemma 2.6. By a similar argument above, the obstructions to lifting the morphism $\mathcal{X}_x^{[1]} \simeq \mathcal{N}^{[1]} \to \mathcal{N}$ to $\mathcal{X}_x^{[n]} \to \mathcal{N}$ vanish as $\operatorname{Ext}^1(\mathbb{L}_{\mathcal{N}/BG_x}, \mathfrak{m}_x^{n+1}/\mathfrak{m}_x^{n+2}) = H^1(BG_x, \Omega^\vee_{\mathcal{N}/BG_x} \otimes \mathcal{J}^{n+1}/\mathcal{J}^{n+2}) = 0$ since $\mathcal{N} \to BG_x$ is smooth. Here we have made the identification $\mathbb{L}_{\mathcal{N}/BG_x} \simeq \Omega_{\mathcal{N}/BG_x}$ by Theorem 2.12(4) and the fact that $\operatorname{Ext}^i(A, B) = H^i(A^\vee \otimes B)$ if A and B are locally free sheaves. Therefore, we obtain isomorphisms $\mathcal{X}_x^{[n]} \simeq \mathcal{N}^{[n]}$ by Lemma 2.7, that is a diagram

$$\begin{tikzcd}
\mathcal{X}_x^{[0]} \arrow[r, hook] & \mathcal{X}_x^{[1]} \arrow[r, hook] & \mathcal{X}_x^{[2]} \arrow[r, hook] & \dots \\
\mathcal{N}^{[0]} \arrow[r, hook] \arrow[u, "\simeq"] & \mathcal{N}^{[1]} \arrow[r, hook] \arrow[u, "\simeq"] & \mathcal{N}^{[2]} \arrow[r, hook] \arrow[u, "\simeq"] & \dots
\end{tikzcd}$$

which can be viewed as an element of $\varprojlim \operatorname{Hom}(\mathcal{N}^{[n]}, \mathcal{X})$.

Since N_x is affine, we can denote $N \coloneqq N_x /\!\!/ G_x$ to be the GIT quotient. Then the morphism $\mathcal{N} \to N$ is a good moduli space as $\mathcal{N}$ is a quotient stack of N_x by G_x which is linearly reductive. Denote by $0 \in N$ the image of the origin. The fibre product $\widehat{\mathcal{N}} \coloneqq \operatorname{Spec} \widehat{\mathcal{O}}_{N,0} \times_N \mathcal{N}$ is a quotient stack of the form $[\operatorname{Spec} B/G]$ where B is of finite type over the complete noetherian local k-algebra $B^G = \widehat{\mathcal{O}}_{N,0}$. We thus get the isomorphisms

$$\begin{aligned}
\varprojlim \operatorname{Hom}(\mathcal{N}^{[n]}, \mathcal{X}) &\simeq \varprojlim \operatorname{Hom}_{r\otimes,\simeq}\Big(\operatorname{Coh}(\mathcal{X}), \operatorname{Coh}(\mathcal{N}^{[n]})\Big) && \text{(Theorem 3.2)}\\
&\simeq \operatorname{Hom}_{r\otimes,\simeq}\Big(\operatorname{Coh}(\mathcal{X}), \varprojlim \operatorname{Coh}(\mathcal{N}^{[n]})\Big) && \text{(Hom preserves limits)}\\
&\simeq \operatorname{Hom}_{r\otimes,\simeq}\Big(\operatorname{Coh}(\mathcal{X}), \operatorname{Coh}(\widehat{\mathcal{N}})\Big) && \text{(Theorem 3.1)}\\
&\simeq \operatorname{Hom}(\widehat{\mathcal{N}}, \mathcal{X}) && \text{(Theorem 3.2).}
\end{aligned}$$

Therefore, there exists a morphism from the legitimate quotient stack $\widehat{\mathcal{N}}$ to $\mathcal{X}$ given by the filling

$$\begin{tikzcd}
\mathcal{X}_x^{[n]} \simeq \mathcal{N}^{[n]} \arrow[r] \arrow[d] & \mathcal{X}. \\
\widehat{\mathcal{N}} \arrow[ur, dashed] &
\end{tikzcd}$$

Lastly, to show that this map is étale, let us consider the contravariant functor

$$F\colon (\mathcal{S}ch/N) \to \mathcal{S}ets, \quad (S \to N) \mapsto \{S \times_N \mathcal{N} \to \mathcal{X}\}/\sim,$$

where $(S \times_N \mathcal{N} \to \mathcal{X}) \sim (S' \times_N \mathcal{N} \to \mathcal{X})$ if there exists a 2-isomorphism between them (which is a groupoid of morphisms). This functor is limit-preserving since $\mathcal{X}$ is of finite type over k. Now the morphism $\widehat{\mathcal{N}} \to \mathcal{X}$ is an element of $F(\operatorname{Spec} \widehat{\mathcal{O}}_{N,0})$. The first part of Theorem 3.5 yields an étale morphism $(U, u) \to (N, 0)$ from a point $u \in U$ in some affine scheme. Let $\mathcal{W} := U \times_N \mathcal{N}$. The second part produces an element in $F(U)$, that is a morphism $f\colon \mathcal{W} \to \mathcal{X}$ such that $f(u, 0) = x$, which agrees with $(\widehat{\mathcal{N}}, 0) \to (\mathcal{X}, x)$ up to first order, i.e. the restrictions of f and $\widehat{\mathcal{N}} \to \mathcal{X}$ to $\operatorname{Spec}(\mathcal{O}_{N,0}/\mathfrak{m}_0^2)$ are equal. This means that $f^{[1]}\colon \mathcal{W}_w^{[1]} \to \mathcal{X}_x^{[1]}$ is an isomorphism (Lemma 2.6) with $w := (u, 0)$. As x is a smooth point, it follows by Lemma 2.7 that f restricts to isomorphisms $f^{[n]}\colon \mathcal{W}_w^{[n]} \to \mathcal{X}_x^{[n]}$ for every n, so f is étale at w. Lastly, $W \simeq [\operatorname{Spec} A/G_x]$ for a finitely generated k-algebra A such that $U = \operatorname{Spec} A^{G_x}$ and $\mathcal{W} \to \mathcal{X}$ induces an isomorphism of stabiliser groups at w). □

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